

RIANA MALHOTRA

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EDUCATION

Bachelor of Applied Science - Computer Engineering + PEY Co-op

Expected May 2029

University of Toronto

- UofT Engineering International Scholar Award Recipient
- Minors: Artificial Intelligence and Engineering Business

TECHNICAL SKILLS

Programming: C++, C, Python, Java, C#, JavaScript, SQL, MATLAB, Verilog

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Multithreading, REST APIs, Debugging, Git

AI/ML: Machine Learning, Generative AI, Computer Vision, OpenCV, PyTorch, NumPy

Systems: Linux/UNIX, QNX RTOS, Embedded Systems, Real-Time Systems

Development: Automation, Integration Testing, Regression Testing, Root-Cause Analysis, Code Reviews, Agile Development

EXPERIENCE

Production Test Engineer Intern | Trimble Appianix

May 2025 - Aug 2025, May 2026 - Aug 2026

- Designed, tested, and debugged 50+ software and firmware builds for airborne and marine GNSS/INS systems using C, C++, Python, QNX, and Linux, resolving software defects and system integration issues before production deployment.
- Developed 15+ C#, C++, and Python automation tools for firmware deployment, configuration, regression testing, and system validation, reducing programming time by approximately 20%.
- Executed automated integration, regression, and system validation tests, analyzing failures and debugging timing, communication, boot, and system-level issues to identify root causes and verify fixes.

Software Member - Avionics Rocketry Subteam | University of Toronto Aerospace Team

Sep 2023 - Present

- Developed embedded C and real-time software for STM32 flight systems, implementing sensor and actuator interfaces and communication protocols for flight-system integration.
- Developed Python automated integration tests to validate communication, timing, data integrity, system behavior, and software interfaces.
- Collaborated with a multidisciplinary engineering team to debug software defects, analyze test results, and validate system requirements during iterative development.

Software Member - Autonomous Rover Team | University of Toronto Robotics Association

Jan 2024 - Dec 2024

- Developed C++ and Python testing tools for autonomous rover systems, validating navigation algorithms, sensor integration, and real-time data processing across simulated and hardware environments.
- Developed Python diagnostic tools to reproduce software failures, analyze system behavior, and accelerate debugging and system validation.

PROJECTS

TransitFlow - GIS Transit Routing Platform | C++, Multithreading, Data Structures & Algorithms, Git

- Designed and developed a C++ GIS application for commuter route planning using A*, Dijkstra, Greedy Search, Ant Colony Optimization, 2-opt, OR-opt, and Simulated Annealing.
- Built a multithreaded optimization pipeline for computationally intensive route planning, improving solution quality by 8% over baseline through algorithm optimization, concurrency, and performance testing.
- Implemented interactive map-based functionality and validated routing correctness, performance, and user scenarios through iterative testing and debugging.
- Collaborated using Git-based development workflows, code reviews, and iterative feature development.

Morse Code Blink Decoder - Computer Vision & FPGA | Python, OpenCV, C++, Verilog, FPGA, UART

- Developed a real-time Python computer vision application using OpenCV to detect and classify eye blinks into Morse-code dot/dash sequences for hands-free text input.
- Designed and implemented a Verilog finite-state machine on an FPGA to parse Morse sequences and decode them into corresponding characters.
- Integrated computer vision with FPGA hardware through UART-to-USB serial communication and implemented VGA output to display decoded text in real time.

TelMe - GenAI Telecom Support Platform | Python, Flask, REST APIs, Machine Learning, Git

- Developed a Python machine learning pipeline using K-Means clustering to segment customers based on usage patterns and support personalized telecom plan recommendations.
- Built an AI-powered conversational application using Python, Flask, React, and REST APIs, integrating machine learning capabilities with application services.
- Applied Git, debugging, testing, and rapid iterative development to deliver a functional software application within a 24-hour hackathon.